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United States Patent	12397727
Kind Code	B2
Date of Patent	August 26, 2025
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Power supply system, and moving object including power supply system

Abstract

A first relay device, a second relay device, a third relay device, a fourth relay device, and a first connection device of a power supply system are disposed in areas different from each other.

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Appl. No.:	18/618014
Filed:	March 27, 2024

Prior Publication Data

Document Identifier	Publication Date
US 20240326727 A1	Oct. 03, 2024

Publication Classification

Int. Cl.: B60R16/03 (20060101); H02J3/38 (20060101); B64D35/021 (20240101)

U.S. Cl.:

CPC B60R16/03 (20130101); H02J3/38 (20130101); B64D35/021 (20240101); B64D2221/00 (20130101)

Field of Classification Search

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
4551632	12/1984	Jourdan et al.	N/A	N/A
2020/0274386	12/2019	Kirleis	N/A	H02J 7/0029
2022/0204173	12/2021	Barraco	N/A	H02J 3/32
2023/0253820	12/2022	Mitani	307/23	B60L 3/04
2023/0312117	12/2022	Tsutsumi	244/17.13	B60L 50/60
2023/0312118	12/2022	Kita	244/17.13	B64D 31/18
2023/0411958	12/2022	Long	N/A	B60L 58/21
2024/0039292	12/2023	Aguilera Medina	N/A	H02J 1/082
2024/0198854	12/2023	Jeon	N/A	B64U 50/34
2024/0199217	12/2023	Saito	N/A	H02J 7/0014
2025/0083572	12/2024	Spagnolo	N/A	H02J 7/34
2025/0096580	12/2024	Saito	N/A	H02J 7/00308

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
6-3975	12/1993	JP	N/A
2020/217007	12/2019	WO	N/A

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application is based upon and claims the benefit of priority from Japanese Patent Application No. 2023-056890 filed on Mar. 31, 2023, the contents of which are incorporated herein by reference.

BACKGROUND OF THE INVENTION

Field of the Invention

(2) The present invention relates to a power supply system, and a moving object including the power supply system.

Description of the Related Art

(3) In recent years, research and development have been conducted on electrification technology that contributes to energy efficiency in order to ensure that more people have access to affordable, reliable, sustainable and modern energy.

(4) JP H06-003975 B discloses a power supply system for an aircraft, which can supply electric power to a load even in an emergency by combining a plurality of generators and storage batteries.

SUMMARY OF THE INVENTION

(5) There has been a demand for a power supply system with more redundancy, and a moving object including such a power supply system.

- (6) The present invention has the object of solving the aforementioned problem.
- (7) A power supply system of the present invention comprises: a first power supply circuit configured to supply, to a first load device, direct current electric power output from a first power generation device; a first power storage device connected to the first power supply circuit in parallel with the first power generation device; a second power supply circuit configured to supply, to a second load device, direct current electric power output from a second power generation device; a second power storage device connected to the second power supply circuit in parallel with the second power generation device; a first relay device including a first partial circuit that is a part of the first power supply circuit, and also including a first contactor device configured to disconnect the first power storage device from the first partial circuit; a second relay device including a second partial circuit that is another part of the first power supply circuit, and also including a second contactor device configured to disconnect the first power generation device from the first partial circuit; a third relay device including a third partial circuit that is a part of the second power supply circuit, and also including a third contactor device configured to disconnect the second power storage device from the third partial circuit; a fourth relay device including a fourth partial circuit that is another part of the second power supply circuit, and also including a fourth contactor device configured to disconnect the second power generation device from the third partial circuit; and a first connection circuit including a first connection device configured to connect the first partial circuit and the third partial circuit to each other, wherein the first relay device, the second relay device, the third relay device, the fourth relay device, and the first connection device are disposed in a plurality of areas different from each other.
- (8) A moving object of the present invention comprises the above-described power supply system.
- (9) According to the present invention, it is possible to provide a power supply system with more redundancy, and a moving object including such a power supply system.
- (10) The above and other objects, features, and advantages of the present invention will become more apparent from the following description when taken in conjunction with the accompanying drawings, in which a preferred embodiment of the present invention is shown by way of illustrative example.
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Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a first schematic view of a power supply system;
- (2) FIG. 2 is a second schematic view of the power supply system;
- (3) FIG. 3 is a diagram showing the operation of the power supply system in a normal state;
- (4) FIG. 4 is a diagram showing the operation of the power supply system when a first abnormality occurs;
- (5) FIG. 5 is a diagram showing the operation of the power supply system when a second abnormality occurs;
- (6) FIG. 6 is a diagram showing the operation of the power supply system when a third abnormality occurs;
- (7) FIG. 7 is a schematic view of a first modification of the power supply system;
- (8) FIG. 8 is a schematic view of a second modification of the power supply system; and
- (9) FIG. 9 is a schematic view of a moving object.

DETAILED DESCRIPTION OF THE INVENTION

- (10) In the power supply system, it is desired that, even if an abnormality occurs in the periphery of one component or in one component itself, the abnormality does not affect the peripheral components. Electric devices are required to have a redundant power supply system that can continue operating even when an abnormality occurs.

First Embodiment

(11) [Configuration of Power Supply System]

(12) A power supply system **10** of a first embodiment will be described with reference to the drawings. FIG. **1** is a first schematic view of the power supply system **10**.

(13) The power supply system **10** includes a first power supply circuit **12a**, a second power supply circuit **12b**, a third power supply circuit **12c**, and a fourth power supply circuit **12d**. The first power supply circuit **12a** supplies, to a first load device **16a**, DC power output from a first power generation device **14a**. The second power supply circuit **12b** supplies, to a second load device **16b**, DC power output from a second power generation device **14b**. The third power supply circuit **12c** supplies, to a third load device **16c**, the DC power output from the first power generation device **14a**. The fourth power supply circuit **12d** supplies, to a fourth load device **16d**, the DC power output from the second power generation device **14b**.

(14) The first power supply circuit **12a** includes a partial circuit **12a2** (a second partial circuit) and a partial circuit **12a1** (a first partial circuit) between the first power generation device **14a** and the first load device **16a**. For example, the partial circuit **12a2** and the partial circuit **12a1** are each provided in a distribution board or a junction box. The partial circuit **12a2** and the partial circuit **12a1** are connected to each other. The partial circuit **12a2** is connected to the first power generation device **14a**. On the other hand, the partial circuit **12a1** is connected to the first load device **16a**. The partial circuit **12a2** and the partial circuit **12a1** relay DC power between the first power generation device **14a** and the first load device **16a**. Hereinafter, a device including the partial circuit **12a2** and a contactor device **22a** (a second contactor device) to be described later is referred to as a relay device **28** (a second relay device). Similarly, a device including the partial circuit **12a1** and a contactor device **26a** (a first contactor device) to be described later is referred to as a relay device **30** (a first relay device).

(15) The second power supply circuit **12b** includes a partial circuit **12b2** (a fourth partial circuit) and a partial circuit **12b1** (a third partial circuit) between the second power generation device **14b** and the second load device **16b**. For example, the partial circuit **12b2** and the partial circuit **12b1** are each provided in a distribution board or a junction box. The partial circuit **12b2** and the partial circuit **12b1** are connected to each other. The partial circuit **12b2** is connected to the second power generation device **14b**. On the other hand, the partial circuit **12b1** is connected to the second load device **16b**. The partial circuit **12b2** and the partial circuit **12b1** relay DC power between the second power generation device **14b** and the second load device **16b**. Hereinafter, a device including the partial circuit **12b2** and a contactor device **22b** (a fourth contactor device) to be described later is referred to as a relay device **36** (a fourth relay device). Similarly, a device including the partial circuit **12b1** and a contactor device **26b** (a third contactor device) to be described later is referred to as a relay device **38** (a third relay device).

(16) The third power supply circuit **12c** includes a partial circuit **12c2** (a seventh partial circuit) and a partial circuit **12c1** (a fifth partial circuit) between the first power generation device **14a** and the third load device **16c**. For example, the partial circuit **12c2** and the partial circuit **12c1** are each provided in a distribution board or a junction box. The partial circuit **12c2** and the partial circuit **12c1** are connected to each other. The partial circuit **12c2** is connected to the first power generation device **14a**. On the other hand, the partial circuit **12c1** is connected to the third load device **16c**. The partial circuit **12c2** and the partial circuit **12c1** relay DC power between the first power generation device **14a** and the third load device **16c**. Hereinafter, a device including the partial circuit **12c1** and a contactor device **26c** (a fifth contactor device) to be described later is referred to as a relay device **34** (a fifth relay device). On the other hand, the partial circuit **12c2** and a contactor device **22c** (a seventh contactor device) to be described later are included in the relay device **28**.

(17) The fourth power supply circuit **12d** includes a partial circuit **12d2** (an eighth partial circuit) and a partial circuit **12d1** (a sixth partial circuit) between the second power generation device **14b** and the fourth load device **16d**. For example, the partial circuit **12d2** and the partial circuit **12d1** are

each provided in a distribution board or a junction box. The partial circuit **12d2** and the partial circuit **12d1** are connected to each other. The partial circuit **12d2** is connected to the second power generation device **14b**. On the other hand, the partial circuit **12d1** is connected to the fourth load device **16d**. The partial circuit **12d2** and the partial circuit **12d1** relay DC power between the second power generation device **14b** and the fourth load device **16d**. Hereinafter, a device including the partial circuit **12d1** and a contactor device **26d** (a sixth contactor device) to be described later is referred to as a relay device **42** (a sixth relay device). On the other hand, the partial circuit **12d2** and a contactor device **22d** (an eighth contactor device) to be described later are included in the relay device **36**.

(18) The first power generation device **14a** and the second power generation device **14b** each include an engine, a generator, and a power control unit (all of them not shown). The engine drives the generator, and the generator generates three-phase AC power. The power control unit converts the three-phase AC power into DC power.

(19) The first power generation device **14a** and the second power generation device **14b** may each include various sensors such as a voltage sensor and a current sensor. Further, the first power generation device **14a** and the second power generation device **14b** may each include elements such as a fuse, a relay, a breaker, a diode, a transistor, a resistor, a coil, and a capacitor.

(20) The first load device **16a**, the second load device **16b**, the third load device **16c**, and the fourth load device **16d** each include an inverter and an electric motor (both of them not shown). The inverter converts the input DC power into three-phase AC power, and the electric motor is driven by the three-phase AC power. The first load device **16a**, the second load device **16b**, the third load device **16c**, and the fourth load device **16d** may each include a DC/DC converter and a low-voltage drive device (both of them not shown). The DC/DC converter lowers the voltage of the input DC power, and the low-voltage drive device is driven by the DC power.

(21) The first load device **16a**, the second load device **16b**, the third load device **16c**, and the fourth load device **16d** may each include various sensors such as a voltage sensor and a current sensor. Further, the first load device **16a**, the second load device **16b**, the third load device **16c**, and the fourth load device **16d** may each include elements such as a fuse, a relay, a breaker, a diode, a transistor, a resistor, a coil, and a capacitor.

(22) The power supply system **10** includes a first connection circuit **18a**. One end of the first connection circuit **18a** is connected to the partial circuit **12a1** of the first power supply circuit **12a**. The other end of the first connection circuit **18a** is connected to the partial circuit **12b1** of the second power supply circuit **12b**. The first connection circuit **18a** includes a first connection device **20a**. The first connection device **20a** can connect the partial circuit **12a1** and the partial circuit **12b1**.

(23) The power supply system **10** includes a second connection circuit **18b**. One end of the second connection circuit **18b** is connected to the partial circuit **12c1** of the third power supply circuit **12c**. The other end of the second connection circuit **18b** is connected to the partial circuit **12d1** of the fourth power supply circuit **12d**. The second connection circuit **18b** includes a second connection device **20b**. The second connection device **20b** can connect the partial circuit **12c1** and the partial circuit **12d1**.

(24) The first connection device **20a** and the second connection device **20b** each include a contactor. The contactor may be configured by a relay or a semiconductor switch. The first connection device **20a** and the second connection device **20b** may each further include a breaker.

(25) Normally, the connection between the first power supply circuit **12a** and the second power supply circuit **12b** is cut off. Thus, when an electrical abnormality occurs in one of the first power supply circuit **12a** or the second power supply circuit **12b**, the other of the first power supply circuit **12a** and the second power supply circuit **12b** can be prevented from being affected by the abnormality. For example, when an overcurrent occurs in one of the first power supply circuit **12a** or the second power supply circuit **12b**, the overcurrent is prevented from flowing to the other of

the first power supply circuit **12a** and the second power supply circuit **12b**.

(26) Similarly, the connection between the third power supply circuit **12c** and the fourth power supply circuit **12d** is normally cut off. Thus, when an electrical abnormality occurs in one of the third power supply circuit **12c** or the fourth power supply circuit **12d**, the other of the third power supply circuit **12c** and the fourth power supply circuit **12d** can be prevented from being affected by the abnormality. For example, when an overcurrent occurs in one of the third power supply circuit **12c** or the fourth power supply circuit **12d**, the overcurrent is prevented from flowing to the other of the third power supply circuit **12c** and the fourth power supply circuit **12d**.

(27) When the supply of electric power from the first power generation device **14a** to the first power supply circuit **12a** and the third power supply circuit **12c** is cut off, the first power supply circuit **12a** and the second power supply circuit **12b** are connected by the first connection device **20a**. Further, the third power supply circuit **12c** and the fourth power supply circuit **12d** are connected by the second connection device **20b**. As a result, electric power is supplied from the second power generation device **14b** to the first power supply circuit **12a** and the third power supply circuit **12c**.

(28) When the supply of electric power from the second power generation device **14b** to the second power supply circuit **12b** and the fourth power supply circuit **12d** is cut off, the first power supply circuit **12a** and the second power supply circuit **12b** are connected by the first connection device **20a**. Further, the third power supply circuit **12c** and the fourth power supply circuit **12d** are connected by the second connection device **20b**. As a result, electric power is supplied from the first power generation device **14a** to the second power supply circuit **12b** and the fourth power supply circuit **12d**.

(29) The power supply system **10** includes the contactor devices **22a** to **22d**. The contactor device **22a** is disposed in the partial circuit **12a2** of the relay device **28**. The contactor device **22a** can disconnect the first power generation device **14a** from the partial circuit **12a1** and the first connection circuit **18a**. The contactor device **22b** is disposed in the partial circuit **12b2** of the relay device **36**. The contactor device **22b** can disconnect the second power generation device **14b** from the partial circuit **12b1** and the first connection circuit **18a**. The contactor device **22c** is disposed in the partial circuit **12c2** of the relay device **28**. The contactor device **22c** can disconnect the first power generation device **14a** from the partial circuit **12c1** and the second connection circuit **18b**. The contactor device **22d** is disposed in the partial circuit **12d2** of the relay device **36**. The contactor device **22d** can disconnect the second power generation device **14b** from the partial circuit **12d1** and the second connection circuit **18b**.

(30) The contactor devices **22a** to **22d** each include a contactor. The contactor may be configured by a relay or a semiconductor switch. The contactor devices **22a** to **22d** may each include a breaker.

(31) The power supply system **10** includes a first power storage device **24a**, a second power storage device **24b**, a third power storage device **24c**, and a fourth power storage device **24d**. The first power storage device **24a** is connected to the first power supply circuit **12a** in parallel with the first power generation device **14a**. The second power storage device **24b** is connected to the second power supply circuit **12b** in parallel with the second power generation device **14b**. The third power storage device **24c** is connected to the third power supply circuit **12c** in parallel with the first power generation device **14a**. The fourth power storage device **24d** is connected to the fourth power supply circuit **12d** in parallel with the second power generation device **14b**.

(32) The first power storage device **24a**, the second power storage device **24b**, the third power storage device **24c**, and the fourth power storage device **24d** each include a lithium ion battery. The first power storage device **24a**, the second power storage device **24b**, the third power storage device **24c**, and the fourth power storage device **24d** may each include a secondary battery other than the lithium ion battery. The first power storage device **24a**, the second power storage device **24b**, the third power storage device **24c**, and the fourth power storage device **24d** may each include a large-capacity capacitor.

(33) The first power storage device **24a**, the second power storage device **24b**, the third power storage device **24c**, and the fourth power storage device **24d** may each include various sensors such as a voltage sensor and a current sensor. The first power storage device **24a**, the second power storage device **24b**, the third power storage device **24c**, and the fourth power storage device **24d** may each include elements such as a fuse, a relay, a breaker, a diode, a transistor, a resistor, a coil, and a capacitor.

(34) The power supply system **10** includes the contactor devices **26a** to **26d**. The contactor device **26a** is included in the relay device **30**. The contactor device **26a** is disposed on a wire connecting the first power storage device **24a** and the partial circuit **12a1**. The contactor device **26a** can disconnect the first power storage device **24a** from the partial circuit **12a1**. The contactor device **26b** is included in the relay device **38**. The contactor device **26b** is disposed on a wire connecting the second power storage device **24b** and the partial circuit **12b1**. The contactor device **26b** can disconnect the second power storage device **24b** from the partial circuit **12b1**. The contactor device **26c** is included in the relay device **34**. The contactor device **26c** is disposed on a wire connecting the third power storage device **24c** and the partial circuit **12c1**. The contactor device **26c** can disconnect the third power storage device **24c** from the partial circuit **12c1**. The contactor device **26d** is included in the relay device **42**. The contactor device **26d** is disposed on a wire connecting the fourth power storage device **24d** and the partial circuit **12d1**. The contactor device **26d** can disconnect the fourth power storage device **24d** from the partial circuit **12d1**.

(35) The contactor devices **26a** to **26d** each include a contactor. The contactor may be configured by a relay or a semiconductor switch. The contactor devices **26a** to **26d** may each include a breaker.

(36) The power supply system **10** may include various sensors such as a voltage sensor and a current sensor, in addition to the above-described configuration. The power supply system **10** may include elements such as a fuse, a resistor, a coil, and a capacitor.

(37) [Arrangement of Components of Power Supply System]

(38) FIG. 2 is a second schematic view of the power supply system **10**. The plurality of components constituting the power supply system **10** are arranged in a distributed manner in a plurality of areas **44**, **46**, **48**, **50**, **52**, **54**, **56**, and **58**. Specifically, the relay device **28**, the relay device **30**, the relay device **34**, the relay device **36**, the relay device **38**, the relay device **42**, the first connection device **20a**, and the second connection device **20b** are disposed in areas different from each other.

(39) The relay device **28** is disposed in an area **44** (a second area). The relay device **30** is disposed in an area **46** (a first area). The relay device **34** is disposed in an area **48** (a sixth area). The relay device **36** is disposed in an area **50** (a fourth area). The relay device **38** is disposed in an area **52** (a third area). The relay device **42** is disposed in an area **54** (a seventh area). The first connection device **20a** (the first connection circuit **18a**) is disposed in an area **56** (a fifth area). The second connection device **20b** (the second connection circuit **18b**) is disposed in an area **58** (an eighth area).

(40) The areas **44**, **46**, **48**, **50**, **52**, **54**, **56**, and **58** are partitioned from each other by partitions **60**. For example, two areas adjacent to each other, such as the area **44** and the area **46** shown in FIG. 2, may be partitioned by a common partition **60**. Further, the areas **44**, **46**, **48**, **50**, **52**, **54**, **56**, and **58** may be partitioned by partitions **60** different from each other. The partition **60** is preferably made of a flame-retardant material, but the material is not limited thereto.

(41) [Operation of Power Supply System in Normal State]

(42) FIG. 3 is a diagram showing the operation of the power supply system **10** in a normal state. Arrows shown in FIG. 3 indicate electric power supply paths.

(43) The first power generation device **14a** is connected to the partial circuit **12a1** and the first connection circuit **18a** by the contactor device **22a**. The first power generation device **14a** is connected to the partial circuit **12c1** and the second connection circuit **18b** by the contactor device **22c**. As a result, electric power is supplied from the first power generation device **14a** to the first

load device **16a** and the third load device **16c**. The second power generation device **14b** is connected to the partial circuit **12b1** and the first connection circuit **18a** by the contactor device **22b**, and the second power generation device **14b** is connected to the partial circuit **12d1** and the second connection circuit **18b** by the contactor device **22d**. As a result, electric power is supplied from the second power generation device **14b** to the second load device **16b** and the fourth load device **16d**.

(44) The first power storage device **24a** is connected to the first load device **16a** by the contactor device **26a**, and electric power is supplied from the first power storage device **24a** to the first load device **16a**. The second power storage device **24b** is connected to the second load device **16b** by the contactor device **26b**, and electric power is supplied from the second power storage device **24b** to the second load device **16b**. The third power storage device **24c** is connected to the third load device **16c** by the contactor device **26c**, and electric power is supplied from the third power storage device **24c** to the third load device **16c**. The fourth power storage device **24d** is connected to the fourth load device **16d** by the contactor device **26d**, and electric power is supplied from the fourth power storage device **24d** to the fourth load device **16d**.

(45) The connection between the partial circuit **12a1** of the first power supply circuit **12a** and the partial circuit **12b1** of the second power supply circuit **12b** is interrupted by the first connection device **20a**. The connection between the partial circuit **12c1** of the third power supply circuit **12c** and the partial circuit **12d1** of the fourth power supply circuit **12d** is interrupted by the second connection device **20b**.

(46) [Operation of Power Supply System when First Abnormality Occurs]

(47) FIG. 4 is a diagram showing the operation of the power supply system **10** when a first abnormality occurs. The arrows indicated by the thick lines in FIG. 4 indicate electric power supply paths. FIG. 4 shows the operation of the power supply system **10** in a case where the supply of electric power from the first power generation device **14a** to the first power supply circuit **12a** and the third power supply circuit **12c** is cut off.

(48) When an abnormality occurs in the area **44** in which the relay device **28** is disposed, the relay device **28** may fail. When the relay device **28** fails, the supply of electric power from the first power generation device **14a** to the first load device **16a** and the third load device **16c** is cut off. In this case, the first power storage device **24a** supplies electric power to the first load device **16a**, and the third power storage device **24c** supplies electric power to the third load device **16c**. However, the first load device **16a** and the third load device **16c** may not be operated for a long time only by the first power storage device **24a** and the third power storage device **24c**.

(49) In the present embodiment, the area **44** is partitioned by the partition **60** shown in FIG. 2. Therefore, the abnormality that has occurred in the area **44** does not affect the components outside the area **44**. Accordingly, the components outside the area **44** can operate normally. Therefore, the electric power supply paths by the remaining power sources can be secured in the power supply system **10**. Consequently, the operation of the power supply system **10** can be continued. According to the present embodiment, it is possible to provide the power supply system **10** with more redundancy.

(50) When an abnormality shown in FIG. 4 occurs, the partial circuit **12a1** of the first power supply circuit **12a** and the partial circuit **12b1** of the second power supply circuit **12b** are connected by the first connection device **20a**. As a result, electric power is supplied from the second power generation device **14b** to the first load device **16a**. Further, the partial circuit **12c1** of the third power supply circuit **12c** and the partial circuit **12d1** of the fourth power supply circuit **12d** are connected by the second connection device **20b**. As a result, electric power is supplied from the second power generation device **14b** to the third load device **16c**. Therefore, the first load device **16a** and the third load device **16c** can be operated.

(51) [Operation of Power Supply System when Second Abnormality Occurs]

(52) FIG. 5 is a diagram showing the operation of the power supply system **10** when a second

abnormality occurs. The arrows indicated by the thick lines in FIG. 5 indicate electric power supply paths. FIG. 5 shows the operation of the power supply system **10** in a case where the supply of electric power from the first power generation device **14a** and the first power storage device **24a** to the first load device **16a** is cut off.

(53) When an abnormality occurs in the area **46** in which the relay device **30** is disposed, the relay device **30** may fail. When the relay device **30** fails, the supply of electric power from the first power generation device **14a** and the first power storage device **24a** to the first load device **16a** is cut off. In this case, the electric power generated by the first power generation device **14a** is used only by the third load device **16c**. Then, the electric power used by the first load device **16a** is left, out of the electric power generated by the first power generation device **14a**.

(54) In the present embodiment, the area **46** is partitioned by the partition **60** shown in FIG. 2. Therefore, the abnormality that has occurred in the area **46** does not affect the components outside the area **46**. Accordingly, the components outside the area **46** can operate normally.

(55) In the case of the abnormality shown in FIG. 5, the partial circuit **12c1** of the third power supply circuit **12c** and the partial circuit **12d1** of the fourth power supply circuit **12d** are connected by the second connection device **20b** disposed in the area **58**. As a result, electric power is supplied from the first power generation device **14a** to the fourth load device **16d**. Therefore, the electric power generated by the first power generation device **14a** is used not only by the third load device **16c** but also by the fourth load device **16d**. Accordingly, the surplus of the electric power generated by the first power generation device **14a** can be effectively utilized.

(56) [Operation of Power Supply System when Third Abnormality Occurs]

(57) FIG. 6 is a diagram showing the operation of the power supply system **10** when a third abnormality occurs. The arrows indicated by the thick lines in FIG. 6 indicate electric power supply paths. FIG. 6 shows the operation of the power supply system **10** when the second connection device **20b** fails.

(58) When an abnormality occurs in the area **58** in which the second connection device **20b** is disposed, the second connection device **20b** may fail. When the second connection device **20b** fails, electric power cannot be supplied from the third power supply circuit **12c** to the fourth power supply circuit **12d**, and electric power cannot be supplied from the fourth power supply circuit **12d** to the third power supply circuit **12c**.

(59) In the present embodiment, the area **58** is partitioned by the partition **60** shown in FIG. 2. Therefore, the abnormality that has occurred in the area **58** does not affect the components outside the area **58**. Accordingly, the components outside the area **58** can operate normally.

(60) When the abnormality shown in FIG. 6 occurs, the power supply system **10** can operate in the same manner as it operates in the normal state shown in FIG. 3.

(61) [Modification]

(62) FIG. 7 is a schematic view of a first modification of the power supply system **10**. As shown in FIG. 7, the first power generation device **14a** may be disposed in the area **44**. The second power generation device **14b** may be disposed in the area **50**. At least one of the first power storage device **24a** or the first load device **16a** may be disposed in the area **46**. At least one of the second power storage device **24b** or the second load device **16b** may be disposed in the area **52**. At least one of the third power storage device **24c** or the third load device **16c** may be disposed in the area **48**. At least one of the fourth power storage device **24d** or the fourth load device **16d** may be disposed in the area **54**.

(63) In this manner, at least one of the first power generation device **14a**, the second power generation device **14b**, the first load device **16a**, the second load device **16b**, the third load device **16c**, the fourth load device **16d**, the first power storage device **24a**, the second power storage device **24b**, the third power storage device **24c**, or the fourth power storage device **24d** may be disposed in the area together with the relay device.

(64) FIG. 8 is a schematic view of a second modification of the power supply system **10**. As shown

in FIG. 8, the areas **44**, **46**, **48**, **50**, **52**, **54**, **56**, and **58** may be spaced apart from each other by a predetermined distance or more. In this case, a gap **62** having the predetermined distance or more is provided between two areas adjacent to each other. The predetermined distance is set in advance so that an abnormality that has occurred in one of two adjacent areas does not affect the other.

(65) In the present embodiment, two load devices and two power storage devices are connected to one power generation device. However, the present invention is not limited thereto. For example, one load device and one power storage device may be connected to one power generation device. Further, three or more load devices and three or more power storage devices may be connected to one power generation device. In addition, the power supply system **10** may include three or more combinations of one power generation device, one or more load devices, and one or more power storage devices.

Second Embodiment

(66) FIG. **9** is a schematic view of a moving object **64**. The power supply system **10** is mounted on the moving object **64**.

(67) The moving object **64** of the present embodiment is an electric vertical take-off and landing aircraft (eVTOL aircraft). The moving object **64** includes eight VTOL rotors **66**. The VTOL rotors **66** generate upward thrust for a fuselage **68**. The moving object **64** includes eight electric motors **70**. One electric motor **70** drives one VTOL rotor **66**. The moving object **64** includes two cruise rotors **72**. The cruise rotors **72** generate forward thrust for the fuselage **68**. The moving object **64** includes four electric motors **74**. Two electric motors **74** drive one cruise rotor **72**.

(68) Each of the first load device **16a**, the second load device **16b**, the third load device **16c**, and the fourth load device **16d** may include two electric motors **70** and one electric motor **74**. Each of the first load device **16a**, the second load device **16b**, the third load device **16c**, and the fourth load device **16d** may include a low-voltage drive device in addition to the electric motors **70** and the electric motor **74**.

(69) The moving object **64** is not limited to an aircraft, and may be a ship, an automobile, a train, or the like.

APPENDICES

(70) The following notes (appendices) are further disclosed in relation to the above-described embodiments.

Appendix 1

(71) The power supply system (**10**) includes: the first power supply circuit (**12a**) configured to supply, to the first load device (**16a**), DC power output from the first power generation device (**14a**); the first power storage device (**24a**) connected to the first power supply circuit in parallel with the first power generation device; the second power supply circuit (**12b**) configured to supply, to the second load device (**16b**), DC power output from the second power generation device (**14b**); the second power storage device (**24b**) connected to the second power supply circuit in parallel with the second power generation device; the first relay device (**30**) including the first partial circuit (**12a1**) that is a part of the first power supply circuit, and also including the first contactor device (**26a**) configured to disconnect the first power storage device from the first partial circuit; the second relay device (**28**) including the second partial circuit (**12a2**) that is another part of the first power supply circuit, and also including the second contactor device (**22a**) configured to disconnect the first power generation device from the first partial circuit; the third relay device (**38**) including the third partial circuit (**12b1**) that is a part of the second power supply circuit, and also including the third contactor device (**26b**) configured to disconnect the second power storage device from the third partial circuit; the fourth relay device (**36**) including the fourth partial circuit (**12b2**) that is another part of the second power supply circuit, and also including the fourth contactor device (**22b**) configured to disconnect the second power generation device from the third partial circuit; and the first connection circuit (**18a**) including the first connection device (**20a**) configured to connect the first partial circuit and the third partial circuit to each other, wherein the

first relay device, the second relay device, the third relay device, the fourth relay device, and the first connection device are disposed in a plurality of areas (**44**, **46**, **50**, **52**, **56**) different from each other.

(72) According to the above configuration, an abnormality that has occurred in a certain area does not affect the components outside the area. Therefore, the components outside the area in which the abnormality has occurred can operate normally. For example, electric power supply paths by the remaining power sources can be secured in the power supply system. That is, according to the above configuration, it is possible to provide the power supply system **10** with more redundancy.

Appendix 2

(73) In the power supply system according to Appendix 1, the plurality of areas may be partitioned from each other by the partition (**60**).

(74) According to the above configuration, two areas adjacent to each other can be reliably separated. Moreover, according to such a configuration, two areas adjacent to each other can be brought close to each other, and the respective components of the power supply system can be disposed in a concentrated manner.

Appendix 3

(75) In the power supply system according to Appendix 1, the plurality of areas may be spaced apart from each other by a predetermined distance or more.

(76) According to the above configuration, since a member for partitioning the areas is not required, the number of components of the power supply system can be reduced.

Appendix 4

(77) In the power supply system according to any one of Appendices 1 to 3, the first relay device may be disposed in the first area (**46**) among the plurality of areas, the second relay device may be disposed in the second area (**44**) among the plurality of areas, the third relay device may be disposed in the third area (**52**) among the plurality of areas, the fourth relay device may be disposed in the fourth area (**50**) among the plurality of areas, the first connection device may be disposed in the fifth area (**56**) among the plurality of areas, the first power generation device may be disposed in the second area (**44**), the second power generation device may be disposed in the fourth area (**50**), the first power storage device and the first load device may be disposed in the first area (**46**), and the second power storage device and the second load device may be disposed in the third area (**52**).

Appendix 5

(78) The power supply system according to any one of Appendices 1 to 3 may further include: the third power supply circuit (**12c**) configured to supply, to the third load device (**16c**), the DC power output from the first power generation device; the third power storage device (**24c**) connected to the third power supply circuit in parallel with the first power generation device; the fourth power supply circuit (**12d**) configured to supply, to the fourth load device (**16d**), the DC power output from the second power generation device; the fourth power storage device (**24d**) connected to the fourth power supply circuit in parallel with the second power generation device; the fifth relay device (**34**) including the fifth partial circuit (**12c1**) that is a part of the third power supply circuit, and also including the fifth contactor device (**26c**) configured to disconnect the third power storage device from the fifth partial circuit; the sixth relay device (**42**) including the sixth partial circuit (**12d1**) that is a part of the fourth power supply circuit, and also including the sixth contactor device (**26d**) configured to disconnect the fourth power storage device from the sixth partial circuit; and the second connection circuit (**18b**) including the second connection device (**20b**) configured to connect the fifth partial circuit and the sixth partial circuit to each other, wherein the second relay device may further include the seventh partial circuit (**12c2**) that is another part of the third power supply circuit, and also include the seventh contactor device (**22c**) configured to disconnect the first power generation device from the fifth partial circuit, the fourth relay device may further include the eighth partial circuit (**12d2**) that is another part of the fourth power supply circuit, and also

include the eighth contactor device (22d) configured to disconnect the second power generation device from the sixth partial circuit, and the first relay device, the second relay device, the third relay device, the fourth relay device, the fifth relay device, the sixth relay device, the first connection device, and the second connection device may be disposed in the areas different from each other.

Appendix 6

(79) In the power supply system according to Appendix 5, the first relay device may be disposed in the first area (46) among the plurality of areas, the second relay device may be disposed in the second area (44) among the plurality of areas, the third relay device may be disposed in the third area (52) among the plurality of areas, the fourth relay device may be disposed in the fourth area (50) among the plurality of areas, the first connection device may be disposed in the fifth area (56) among the plurality of areas, the fifth relay device may be disposed in the sixth area (48) among the plurality of areas, the sixth relay device may be disposed in the seventh area (54) among the plurality of areas, the second connection device may be disposed in the eighth area (58) among the plurality of areas, the first power generation device may be disposed in the second area (44), the second power generation device may be disposed in the fourth area (50), the first power storage device and the first load device may be disposed in the first area (46), the second power storage device and the second load device may be disposed in the third area (52), the third power storage device and the third load device may be disposed in the sixth area (48), and the fourth power storage device and the fourth load device may be disposed in the seventh area (54).

Appendix 7

(80) The moving object (64) includes the power supply system according to any one of Appendices 1 to 6.

(81) The present invention is not limited to the above disclosure, and various modifications are possible without departing from the essence and gist of the present invention.

Claims

1. A power supply system comprising: a first power supply circuit configured to supply, to a first load device, direct current electric power output from a first power generation device; a first power storage device connected to the first power supply circuit in parallel with the first power generation device; a second power supply circuit configured to supply, to a second load device, direct current electric power output from a second power generation device; a second power storage device connected to the second power supply circuit in parallel with the second power generation device; a first relay device including a first partial circuit that is a part of the first power supply circuit, and also including a first contactor device configured to disconnect the first power storage device from the first partial circuit; a second relay device including a second partial circuit that is another part of the first power supply circuit, and also including a second contactor device configured to disconnect the first power generation device from the first partial circuit; a third relay device including a third partial circuit that is a part of the second power supply circuit, and also including a third contactor device configured to disconnect the second power storage device from the third partial circuit; a fourth relay device including a fourth partial circuit that is another part of the second power supply circuit, and also including a fourth contactor device configured to disconnect the second power generation device from the third partial circuit; and a first connection circuit including a first connection device configured to connect the first partial circuit and the third partial circuit to each other, wherein the first relay device, the second relay device, the third relay device, the fourth relay device, and the first connection device are disposed in a plurality of areas different from each other.

2. The power supply system according to claim 1, wherein the plurality of areas are partitioned from each other by a partition.

3. The power supply system according to claim 1, wherein the plurality of areas are spaced apart from each other by a predetermined distance or more.

4. The power supply system according to claim 1, wherein the first relay device is disposed in a first area among the plurality of areas, the second relay device is disposed in a second area among the plurality of areas, the third relay device is disposed in a third area among the plurality of areas, the fourth relay device is disposed in a fourth area among the plurality of areas, the first connection device is disposed in a fifth area among the plurality of areas, the first power generation device is disposed in the second area, the second power generation device is disposed in the fourth area, the first power storage device and the first load device are disposed in the first area, and the second power storage device and the second load device are disposed in the third area.

5. The power supply system according to claim 1, further comprising: a third power supply circuit configured to supply, to a third load device, the direct current electric power output from the first power generation device; a third power storage device connected to the third power supply circuit in parallel with the first power generation device; a fourth power supply circuit configured to supply, to a fourth load device, the direct current electric power output from the second power generation device; a fourth power storage device connected to the fourth power supply circuit in parallel with the second power generation device; a fifth relay device including a fifth partial circuit that is a part of the third power supply circuit, and also including a fifth contactor device configured to disconnect the third power storage device from the fifth partial circuit; a sixth relay device including a sixth partial circuit that is a part of the fourth power supply circuit, and also including a sixth contactor device configured to disconnect the fourth power storage device from the sixth partial circuit; and a second connection circuit including a second connection device configured to connect the fifth partial circuit and the sixth partial circuit to each other, wherein the second relay device further includes a seventh partial circuit that is another part of the third power supply circuit, and also includes a seventh contactor device configured to disconnect the first power generation device from the fifth partial circuit, the fourth relay device further includes an eighth partial circuit that is another part of the fourth power supply circuit, and also includes an eighth contactor device configured to disconnect the second power generation device from the sixth partial circuit, and the first relay device, the second relay device, the third relay device, the fourth relay device, the fifth relay device, the sixth relay device, the first connection device, and the second connection device are disposed in the areas different from each other.

6. The power supply system according to claim 5, wherein the first relay device is disposed in a first area among the plurality of areas, the second relay device is disposed in a second area among the plurality of areas, the third relay device is disposed in a third area among the plurality of areas, the fourth relay device is disposed in a fourth area among the plurality of areas, the first connection device is disposed in a fifth area among the plurality of areas, the fifth relay device is disposed in a sixth area among the plurality of areas, the sixth relay device is disposed in a seventh area among the plurality of areas, the second connection device is disposed in an eighth area among the plurality of areas, the first power generation device is disposed in the second area, the second power generation device is disposed in the fourth area, the first power storage device and the first load device are disposed in the first area, the second power storage device and the second load device are disposed in the third area, the third power storage device and the third load device are disposed in the sixth area, and the fourth power storage device and the fourth load device are disposed in the seventh area.

7. A moving object comprising the power supply system according to claim 1.
